

EX01

The screenshot shows the pgAdmin 4 interface. On the left, the 'Servers' tree is expanded to 'PostgreSQL 18' > 'Databases (3)' > 'GAME-ARENA' > 'biblioteca'. The 'Query' tab is active, displaying the following SQL query:

```
SELECT
  a.id_autor,
  a.nome_autor,
  (
    SELECT COUNT(*)
    FROM livro l
    WHERE l.id_autor = a.id_autor
  ) AS qtd_livros,
  (
    SELECT ROUND(AVG(l2.paginas), 2)
    FROM livro l2
    WHERE l2.id_autor = a.id_autor
  ) AS media_paginas
FROM autor a
ORDER BY a.id_autor;
```

The 'Data Output' tab shows the results of the query:

id_autor [PK] integer	nome_autor character varying (100)	qtd_livros bigint	media_paginas numeric
1	George Orwell	2	240.00
2	J.K. Rowling	1	264.00
3	Stephen King	2	784.00

Summary: Total rows: 3, Query complete 00:00:00.142. The status bar shows 'CRLF Ln 96, Col 21'.

The screenshot shows the pgAdmin 4 interface with a more complex SQL query using a Common Table Expression (CTE):

```
-- CTE |
WITH estatisticas AS (
  SELECT
    l.id_autor,
    COUNT(*) AS qtd_livros,
    ROUND(AVG(l.paginas), 2) AS media_paginas
  FROM livro l
  GROUP BY l.id_autor
)
SELECT
  a.id_autor,
  a.nome_autor,
  COALESCE(e.qtd_livros, 0) AS qtd_livros,
  e.media_paginas
FROM autor a
LEFT JOIN estatisticas e ON e.id_autor = a.id_autor
ORDER BY a.id_autor;
```

The 'Data Output' tab shows the results:

id_autor [PK] integer	nome_autor character varying (100)	qtd_livros bigint	media_paginas numeric
1	George Orwell	2	240.00
2	J.K. Rowling	1	264.00
3	Stephen King	2	784.00

Summary: Total rows: 3, Query complete 00:00:00.210. The status bar shows 'CRLF Ln 98, Col 8'.

EX02

Servers (1)
PostgreSQL 18
Databases (3)
GAME_ARENA
biblioteca
Casts
Catalogs
Event Triggers
Extensions
Foreign Data W
Languages
Publications
Schemas
Subscriptions
postgres
Login/Group Roles
Tablespaces

Query
Query History

117 -- EX02
118 WITH paginas_por_autor AS (
119 SELECT
120 a.id_autor,
121 a.nome_autor,
122 SUM(l.paginas) AS total_paginas
123 FROM autor a
124 JOIN livro l ON l.id_autor = a.id_autor
125 GROUP BY a.id_autor, a.nome_autor
126)
127 SELECT
128 p.id_autor,
129 p.nome_autor,
130 p.total_paginas
131 FROM paginas_por_autor p
132 WHERE p.total_paginas > (
133 SELECT AVG(total_paginas)
134 FROM paginas_por_autor
135)
136 ORDER BY p.total_paginas DESC;

Data Output
Messages
Notifications

Showing rows: 1 to 1
Page No: 1 of 1
CRLF
Ln 129, Col 15

id_autor	nome_autor	total_paginas
[PK] integer	character varying (100)	bigint
1	3 Stephen King	1568

Total rows: 1
Query complete 00:00:00.160

21°C
Nublado

Pesquisar

11:50
18/05/2026

EX03

```
145 (
146     SELECT SUM(l1.paginas)
147     FROM livro l1
148     WHERE l1.id_autor = a.id_autor
149 ) AS total_paginas
150 FROM autor a
151 WHERE (
152     SELECT SUM(l2.paginas)
153     FROM livro l2
154     WHERE l2.id_autor = a.id_autor
155 ) > (
156     SELECT AVG(x.total_paginas)
157     FROM (
158         SELECT SUM(l3.paginas) AS total_paginas
159         FROM livro l3
160         GROUP BY l3.id_autor
161     ) AS x
162 )
163 ORDER BY total_paginas DESC;
```

id_autor	nome_autor	total_paginas
3	Stephen King	1568

Total rows: 1 Query complete 00:00:00.189

```
167 WITH paginas_por_autor AS (
168     SELECT
169         a.id_autor,
170         a.nome_autor,
171         SUM(l.paginas) AS total_paginas
172     FROM autor a
173     JOIN livro l ON l.id_autor = a.id_autor
174     GROUP BY a.id_autor, a.nome_autor
175 )
176 SELECT
177     p.id_autor,
178     p.nome_autor,
179     p.total_paginas
180 FROM paginas_por_autor p
181 WHERE p.total_paginas > (
182     SELECT AVG(total_paginas)
183     FROM paginas_por_autor
184 )
185 ORDER BY p.total_paginas DESC;
```

id_autor	nome_autor	total_paginas
3	Stephen King	1568

Total rows: 1 Query complete 00:00:00.130

ALTER TABLE livro

ADD COLUMN paginas INT;

UPDATE livro SET paginas = 328 WHERE id_livro = 1;

UPDATE livro SET paginas = 152 WHERE id_livro = 2;

UPDATE livro SET paginas = 264 WHERE id_livro = 3;

```
UPDATE livro SET paginas = 464 WHERE id_livro = 4;
```

```
UPDATE livro SET paginas = 1104 WHERE id_livro = 5;
```

```
-- EX01
```

```
SELECT
```

```
    a.id_autor,
```

```
    a.nome_autor,
```

```
    (
```

```
        SELECT COUNT(*)
```

```
        FROM livro l
```

```
        WHERE l.id_autor = a.id_autor
```

```
    ) AS qtd_livros,
```

```
    (
```

```
        SELECT ROUND(AVG(l2.paginas), 2)
```

```
        FROM livro l2
```

```
        WHERE l2.id_autor = a.id_autor
```

```
    ) AS media_paginas
```

```
FROM autor a
```

```
ORDER BY a.id_autor;
```

```
-- CTE
```

```
WITH estatisticas AS (
```

```
    SELECT
```

```
        l.id_autor,
```

```
        COUNT(*) AS qtd_livros,
```

```
        ROUND(AVG(l.paginas), 2) AS media_paginas
```

```
    FROM livro l
```

```
    GROUP BY l.id_autor
```

```
)  
  
SELECT  
  
    a.id_autor,  
  
    a.nome_autor,  
  
    COALESCE(e.qtd_livros, 0) AS qtd_livros,  
  
    e.media_paginas  
  
FROM autor a  
  
LEFT JOIN estatisticas e ON e.id_autor = a.id_autor  
  
ORDER BY a.id_autor;
```

-- EX02

```
WITH paginas_por_autor AS (  
  
    SELECT  
  
        a.id_autor,  
  
        a.nome_autor,  
  
        SUM(l.paginas) AS total_paginas  
  
    FROM autor a  
  
    JOIN livro l ON l.id_autor = a.id_autor  
  
    GROUP BY a.id_autor, a.nome_autor  
  
)
```

```
SELECT  
  
    p.id_autor,  
  
    p.nome_autor,  
  
    p.total_paginas  
  
FROM paginas_por_autor p  
  
WHERE p.total_paginas > (  
  
    SELECT AVG(total_paginas)  
  
    FROM paginas_por_autor
```

)

ORDER BY p.total_paginas DESC;

-- EX03

--(A) Versão subconsulta correlacionada

SELECT

 a.id_autor,

 a.nome_autor,

 (

 SELECT SUM(l1.paginas)

 FROM livro l1

 WHERE l1.id_autor = a.id_autor

) AS total_paginas

FROM autor a

WHERE (

 SELECT SUM(l2.paginas)

 FROM livro l2

 WHERE l2.id_autor = a.id_autor

) > (

 SELECT AVG(x.total_paginas)

 FROM (

 SELECT SUM(l3.paginas) AS total_paginas

 FROM livro l3

 GROUP BY l3.id_autor

) AS x

)

```
ORDER BY total_paginas DESC;
```

```
--(B) Versão CTE
```

```
WITH paginas_por_autor AS (  
    SELECT  
        a.id_autor,  
        a.nome_autor,  
        SUM(l.paginas) AS total_paginas  
    FROM autor a  
    JOIN livro l ON l.id_autor = a.id_autor  
    GROUP BY a.id_autor, a.nome_autor  
)  
  
SELECT  
    p.id_autor,  
    p.nome_autor,  
    p.total_paginas  
FROM paginas_por_autor p  
WHERE p.total_paginas > (  
    SELECT AVG(total_paginas)  
    FROM paginas_por_autor  
)  
  
ORDER BY p.total_paginas DESC;
```